

Zain Rizvi

647-929-4110 | rizvi-zain@hotmail.com | [linkedin.com/in/zain-rizvi-69b175215/](https://www.linkedin.com/in/zain-rizvi-69b175215/) | github.com/ZainRizvi9

EDUCATION

University of Ottawa

Honours BSc Computer Science Co-Op

Ottawa, ON

Sept. 2023 – Dec 2027

PROJECTS

Cosmora | React, Three.js, React Three Fiber, NASA API, Vercel

- cosmora-mu.vercel.app
- Demo: <https://youtu.be/gGv5bNrwf8Y>
- Built a 3D interactive solar system explorer with real-time free-fly camera navigation and NASA texture-mapped planets
- Implemented Kepler's orbital equations to compute accurate planetary positions for any historical or future date
- Integrated live NASA NeoWs API for near-Earth asteroid data and wheretheiss.at API for real-time ISS tracking
- Built a K-nearest-neighbours threat classifier from scratch in vanilla JS to predict asteroid danger level from size, velocity, and miss distance
- Engineered a planetary alignment detector using angular position computation across all 8 planets

FilmFlicker | Python, Streamlit, Pandas, Plotly, TMDB API

- filmflicker.streamlit.app
- Demo: https://youtu.be/OVAbEVROX_w
- Built a movie recommendation engine using TMDB's graph-based expansion API with vote threshold and release year filtering
- Engineered genre one-hot encoding and normalized feature vectors (rating, year, popularity) to compute weighted cosine similarity matrices across user-selected films
- Implemented a genre-based KNN vibe classifier that maps user taste profiles to curated film categories across 15 hand-labeled lists
- Built interactive data visualizations including a genre donut chart, pick-order vs rating scatter plot, and film similarity heatmap
- Integrated TMDB trending API to dynamically refresh the movie pool each session alongside 300+ curated titles across 15 vibe categories

Portfolio Website | React, JavaScript, AWS EC2, Nginx, CI/CD

- zainrizvi9.github.io
- Built an automated CI/CD pipeline using AWS EC2, Nginx, and GitHub webhooks

EXPERIENCE

IT Service Data Scientist Co-op

York Region

May 2024 – Aug 2024, May 2025 – Aug 2025

Newmarket, ON

- Queried and cleaned ticketing system data using SQL/Excel to support weekly and monthly reporting
- Built simple dashboards and charts (incident volume, categories, resolution times) to track trends over time
- Produced summary reports for the team by grouping incidents by service, priority, and time period
- Helped maintain consistent data entry by standardizing labels/tags and documenting common issue categories
- Worked with technicians to verify numbers, clarify ticket details, and update documentation for recurring issues

Director of Academic Affairs

Computer Science Student Association (CSSA)

Sept. 2023 – May 2024

University of Ottawa

- Organized professional development events including a Nokia office tour for 50+ students
- Led academic outreach initiatives to strengthen student-faculty collaboration

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Java, HTML/CSS, R

Data & Visualization: Pandas, Plotly, Streamlit, Recharts, EDA, API Integration

Frameworks: React, Node.js, Express, Three.js, Flask, FastAPI

Tools: Git, AWS, Docker, VS Code, PostgreSQL, MongoDB

COURSEWORK

Relevant: Data Structures & Algorithms, Databases, Statistics, Object-Oriented Programming, Computer Architecture